

Real Estate Development as an Option

JOSEPH T. WILLIAMS

*Professor of Finance and Urban Land Economics, Faculty of Commerce and Business Administration,
University of British Columbia, 2053 Main Mall, Vancouver, British Columbia V6T 1Z2*

Abstract

Subject to legal limitations, the owner of undeveloped real estate can determine both the date and density at which to develop his property. Alternatively, he can abandon his property. The value of these options depends partly on the stochastic evolution through time of the operating revenues and construction costs of developed property. In this paper the option pricing problem is solved analytically and numerically for the optimal data and density of development, the optimal date of abandonment, and the resulting market values of the developed and undeveloped properties.

Keywords: Development, option.

The option to develop real estate differs in several important ways from put and call options that investors can trade on organized exchanges. Subject to legal limitations, the owner of real estate can select the scale or density at which to develop his property. Also, if the costs of carrying an undeveloped property exceed sufficiently its operating revenues, then the owner has an incentive to abandon his property. Both options affect the market value of the undeveloped property. In addition, the value of the undeveloped property depends upon not only the operating revenues from the developed property but also its costs of development, both of which can evolve stochastically over time.¹ Compared to problems with financial options, the owner's options are more complex and are driven by a different set of stochastic state variables.

In other ways developing real estate is much like exercising an option. Subject to local zoning regulations, the owner chooses the date at which to develop his property. This option is more valuable the more uncertain are changes over time in either operating revenues or construction costs. If investors can trade substitute securities continuously without transaction costs in a perfectly competitive capital market, then the owner optimally develops his property only when development maximizes its market value. For example, maximizing market value is optimal if the capital market is perfectly competitive and complete, even if the market for real estate is not. Also, since the option to develop never expires, the owner optimally exercises his option only when his property's developed value, as determined by its operating cash inflows, exceeds its costs of development. Depending upon the values of various parameters, the difference at the optimal exercise point between the developed value and the construction costs can be considerable. Finally, development is essentially irrevocable, much like the exercise of financial options. Because buildings are durable, properties are redeveloped only after prior improvements are rendered economically or technically obsolete by the passage of time.

Previous papers on this problem include Titman (1985) and Capozza and Sick (1988).² In Titman's article the development of urban property is recast as a problem in option pricing and then solved using standard techniques. The results have interesting implications for both the price of urban land and the partial development of urban property, explaining why, for example, high-rise buildings sometimes abut parking lots. Because the option to build is valuable, development is optimally deferred until the value of developed property is greater than its construction cost and, possibly, much greater. In Capozza and Sick's paper, landowners have the option to convert agricultural land into urban land. Urban land is priced in a competitive, complete capital market, and the price of agricultural land is then calculated analytically, using standard techniques in option pricing.³ The optimal point for converting agricultural into urban land is expressed in terms of the distance from the city center, using a model of the demand for urban land in which urban rents are linear in the distance. The results have interesting implications for both the boundary of the urban area and the income multiplier of agricultural land.

In this paper real estate is valued as an option, and the optimal points at which to abandon and to develop property, as well as the optimal density of development are determined. By assumption, both the operating cash inflows from the developed property and its costs of development are driven by geometric Wiener processes; the cost of density is Cobb-Douglas; and the maximum feasible density is determined by zoning restrictions.⁴ In the initial model undeveloped property has no carrying cost. This initial problem is presented in Section 1 and solved in Section 2. The analytical solution includes the optimal point and density of development and the values of developed and undeveloped property. A stochastic carrying cost for the undeveloped property is then added in Section 3. In this second situation the option to abandon the property is valuable, and the solution must be computed numerically. From the perspective of the literature on option pricing, the novel results relate to the optimal and maximum densities of development. Finally, the main results are summarized in Section 4. All technical details are relegated to the Appendix.

1. Initial model

An investor owns an undeveloped or underdeveloped property. Subject to legal limitations, he can choose both the date and density at which to develop his property. The date at which the owner acquired his property is denoted by $t = 0$. At any time $t \geq 0$, he can develop his property at some feasible density, scale, or quantity q satisfying $1 \leq q \leq \delta$, where δ denotes the maximum density permitted under zoning regulations. Development is costly. The cost, $q^\gamma x_1$, depends on the development cost per unit of density x_1 , measured per unit of time, and the constant cost of scale γ . Development at higher densities is more costly per unit of development: $\gamma > 1$. A property developed at the density q produces the net cash inflow qx_2 per unit of time. With this technology x_2 is the cash inflow per unit of density, also measured per unit of time. By contrast, the undeveloped property has the corresponding net cash inflow per unit of time, βx_2 . With the constant $0 \leq \beta < 1$ and the subsequent specification of the variable x_1 , development always increases the property's net cash inflow.

Both the development cost and the net cash inflows evolve stochastically through time. By assumption, the unit development cost x_1 and the unit cash inflows x_2 follow geometric Wiener processes:⁵

$$dx_i = \mu_i x_i dt + \sigma_i x_i dz_i, \quad (1)$$

where $i = 1, 2$. Each variable x_i has the constant, expected rate of growth μ_i and the constant variance of the growth rate σ_i^2 , both measured per unit of time. Also, the covariance σ_{12} between the rates of growth in x_1 and x_2 is constant per unit of time. In this case the correlation coefficient, $\rho \equiv \sigma_{12}/\sigma_1\sigma_2$, is also constant. With this specification the cash inflows from the undeveloped property are almost always positive. Potentially negative cash inflows are introduced in Section 3.⁶

To value the undeveloped and developed properties, some additional assumptions are necessary. The riskless rate of interest ι is constant per unit of time. Also, the stochastic evolution (1) of both the unit construction cost x_1 and the unit cash inflow x_2 can be replicated from portfolios of securities that are traded continuously without transactions costs in the perfectly competitive capital market. In this market there are two portfolios of traded securities, $i = 1, 2$, such that the returns on portfolio i are perfectly locally correlated with the stochastic increments, dz_i , in (1). For each portfolio the excess mean return per unit of standard deviation equals some constant λ_i . In this case, the risk-adjusted, expected growth rates: $\nu_i \equiv \mu_i - \lambda_i\sigma_i$, $i = 1, 2$, are also constant.⁷ Assuming that the riskless rate of interest ι satisfies the inequalities, $\nu_2 < \iota \leq 1 + \nu_2$, then the developed and undeveloped properties have finite values, as computed below.

Focus first on the developed property. Over time the value of the developed property evolves in response to the stochastic evolution of its net cash inflow. Conditional on the current unit cash inflow x_2 , the developed property has the current price $P(x_2)$. Because the price $P(x_2)$ is dependent only on the net cash inflow x_2 and various parameters, an instantaneously riskless portfolio can be constructed from the developed property and the second portfolio of substitute securities, $i = 2$. To preclude riskless arbitrage, this instantaneously riskless portfolio must yield the riskless rate ι . Following a now familiar argument in the literature on option pricing, the price $P(x_2)$ of the developed property must satisfy the valuation equation:⁸

$$0 = \frac{1}{2} \sigma_2^2 x_2^2 P'' + \nu_2 x_2 P' - \iota P + q x_2, \quad (2)$$

with the density of development q . Because development is irreversible, the differential equation (2) must be satisfied for all feasible net cash inflows: $x_2 \geq 0$.

The price of the developed property must also satisfy two boundary conditions. If the developed property has no net cash inflow, $x_1 = 0$, then, with the geometric Wiener process (1), it has no net cash inflow from that time forward. Thus, if $x_1 = 0$, then the developed property is worthless:

$$P(0) = 0. \quad (3)$$

Also, if the property is to have a well-defined income multiplier, then its price per unit of cash inflow, $P(x_2)/x_2$, must be bounded above by some constant, $0 < \xi < \infty$:

$$P(x_2) \leq \xi x_2. \quad (4)$$

The valuation equation (2) and boundary conditions, (3) and (4), uniquely determine the pricing function, P , for the developed property.

Over time the value of the undeveloped property is driven by the random evolution of both the unit construction cost and the unit cash inflow. Conditional on the current values, $x = (x_1, x_2)$, the undeveloped property has the value $V(x)$. Given the two state variables x , an instantaneously riskless portfolio can be constructed by combining the undeveloped property with two portfolios of substitute securities, $i = 1, 2$. Again, the returns on the two portfolios are perfectly locally correlated with the stochastic increments in (1). To preclude riskless arbitrage, the riskless portfolio must then yield the riskless rate of return ι . By a familiar argument in option pricing, the value of the undeveloped property V must then satisfy the valuation equation:⁹

$$0 = \frac{1}{2} \sigma_1^2 x_1^2 V_{11} + \sigma_{12} x_1 x_2 V_{12} + \frac{1}{2} \sigma_2^2 x_2^2 V_{22} + \nu_1 x_1 V_1 + \nu_2 x_2 V_2 - \iota V + \beta x_2. \quad (5)$$

In (5) βx_2 is the net cash inflow per unit of time from the undeveloped property. This partial differential equation must be satisfied at all values x for which development of the property is not optimal.

The value of the undeveloped property must also satisfy boundary conditions. The first boundary condition is determined as follows. Given the stochastic specification (1), neither the unit development cost x_1 nor the unit cash inflow x_2 can be negative. As a result, the values of the developed and undeveloped properties must satisfy the inequalities: $0 \leq V(x) \leq P(x_2)$. In addition, if the developed property has no current cash inflow, $x_2 = 0$, then it is worthless, as specified in (3). In this case, the undeveloped property must also be worthless:

$$V(x_1, 0) = 0. \quad (6)$$

Under the above assumptions, the investor picks both the optimal point and optimal density of development that maximizes the market value of his undeveloped property. If the property is developed at the point $x = x^*$ with the density $q = q^*$, then its value $V(x^*)$ must be equal the price of the developed property $P(x_2^*)$ minus the cost of development $q^* \gamma x_1^*$:

$$V(x^*) = P(x_2^*) - q^* \gamma x_1^*. \quad (7)$$

Given (7), development is optimal when $x = x^*$ at $q = q^*$ if and only if

$$V_1(x^*) = -q^*, \quad V_2(x^*) = P'(x_2^*), \quad (8)$$

and

$$q^* \equiv \arg \max_q \{P(x_2^*) - q^\gamma x_1^*: 1 \leq q \leq \delta\}. \quad (9)$$

Condition (8) is the tight-fit or smooth-pasting condition familiar from option pricing.¹⁰ These optimality conditions are verified in the Appendix.

2. Optimal development

Problem (2) through (9) is solved in two steps. First, the pricing function P for the developed property is determined. Next, using this result, a valuation function V for the undeveloped property is derived. For the developed property the solution is immediate. Subject to the boundary conditions (3) and (4), the differential equation (2) has the unique solution:

$$P(x_2) = \pi q x_2. \quad (10)$$

with the constant $\pi \equiv 1/(\iota - \nu_2)$. As indicated in (10), the price of the developed property is linear in its net cash inflow per unit of time $q x_2$. The price per dollar of cash inflow, or, equivalently, the income multiplier π is decreasing in the interest rate ι , increasing in the expected growth rate μ_2 of the unit cash inflow x_2 , and decreasing in the associated variance σ_2^2 .

Given (10), the value of the undeveloped property can be calculated as follows. Equations (5) through (10) are satisfied by a valuation function V that is linear homogeneous in x . Accordingly, two ratios are defined: the unit cash inflow relative to the unit construction cost, $y \equiv x_2/x_1$, and the value of the undeveloped property relative to the unit construction cost, $W(y) \equiv V(x)/x_1$. Also, two parameters are defined:

$$\eta \equiv - \left[\frac{\nu_2 - \nu_1}{\omega^2} - \frac{1}{2} \right] + \sqrt{\left[\frac{\nu_2 - \nu_1}{\omega^2} - \frac{1}{2} \right]^2 + 2 \frac{\iota - \nu_1}{\omega^2}}, \quad \psi \equiv \frac{\eta}{\eta - 1},$$

with $\omega^2 \equiv \sigma_1^2 - 2\sigma_{12} + \sigma_2^2$. These parameters satisfy the inequalities: $\eta, \psi > 1$.¹¹ With this transformation problem (5) through (10) simplifies as shown in the Appendix. This simplified problem has the unique solution specified below.

The solution includes both the optimal development point and optimal density of development. Development optimally occurs at the ratio y^* :

$$y^* = \begin{cases} \frac{\psi}{\pi} \frac{1}{1 - \beta\psi} & \frac{\psi}{1 - \beta\psi} \leq \gamma \\ \frac{\gamma}{\pi} \left(\frac{\beta\gamma\psi}{\gamma - \psi} \right)^{\gamma-1} & \frac{\delta\psi}{\delta - \beta\psi} \leq \gamma < \frac{\psi}{1 - \beta\psi} \\ \frac{\psi}{\pi} \frac{\delta\gamma}{\delta - \beta\psi} & \gamma < \frac{\delta\psi}{\delta - \beta\psi} \end{cases} \quad (11)$$

Also, the optimal density of development q^* is¹²

$$q^* = \begin{cases} 1 & \frac{\psi}{1 - \beta\psi} \leq \gamma \\ \frac{\beta\gamma\psi}{\gamma - \psi} & \frac{\delta\psi}{\delta - \beta\psi} \leq \gamma < \frac{\psi}{1 - \beta\psi} \\ \delta & \gamma < \frac{\delta\psi}{\delta - \beta\psi} \end{cases} \quad (12)$$

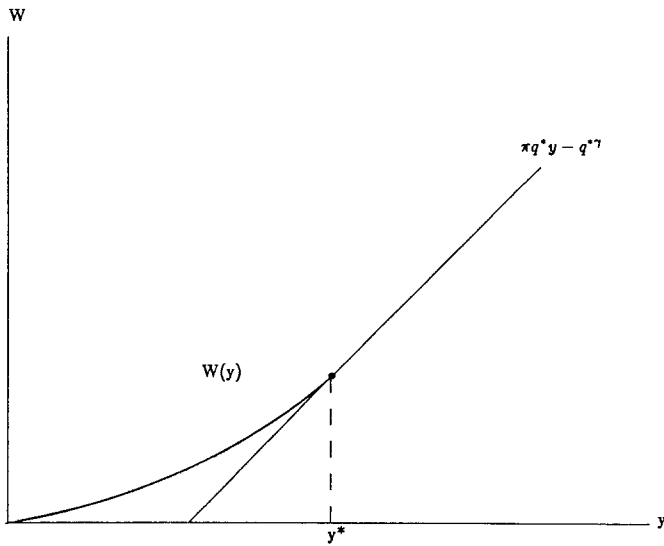
The resulting value of the undeveloped property can be written conditional on the optimal ratio and density of development as follows. Measured per unit of the development cost, $W \equiv V/x_1$, the value of the undeveloped property is

$$W(y) = \beta\pi y + \frac{q^*}{\eta - 1} \left(\frac{y}{y^*} \right)^\eta, \quad (13)$$

for $0 \leq y \leq y^*$. This solution is depicted in Figure 1. As indicated, the undeveloped property's value relative to its construction cost (13) is increasing and convex in the ratio $y \equiv z_2/x_1$ and tangent at the optimal development ratio y^* to the development profit per unit of development cost, $\pi q^* y - q^{*\gamma}$. As a result, the income multiplier, $V/x_2 = W/y$, is increasing in the ratio y of the unit cash inflow relative to the unit construction cost.

This solution has the following comparative statics. An increase in either variance, σ_1^2 or σ_2^2 , or a decrease in the correlation ρ that does not change the risk-adjusted, expected growth rates, ν_1 and ν_2 , increases the development ratio (11), density (12), and value (13), whenever $\nu_1 \geq \nu_2$. Under the same conditions, an increase in the riskless rate of interest ι has the opposite effect; it decreases (11) through (13). An increase in the cash inflow from the undeveloped property βx_2 , induced by an increase in the coefficient β , increases both the optimal ratio (11) and the optimal density (12). Whenever the maximum density constrains development, $q^* = \delta$, an increase in the maximum density δ decreases the development ratio (11) but increases both the optimal density (12) and the optimal ratio (13). Finally, an increase in the cost of scale γ decreases the optimal density (12) whenever $0 < q^* < \delta$. All other comparative statics depend, in a complicated fashion, on the values of the various parameters. The results are summarized in Table 1.

Of these comparative statics, only the parts played by the optimal and maximum densities are novel to the literature on option pricing. Consider first the comparative statics for the optimal density. Whenever the risk-adjusted, expected growth rate of construction costs exceeds or equals the corresponding rate for operating cash inflows, $\nu_1 \geq \nu_2$, development optimally occurs at higher densities (12) with either more risky properties or lower interest rates. These results can be understood as follows. The option to develop is more valuable with a flexible than a fixed density. Given a flexible density, the owner then optimally defers development until he realizes a higher ratio of rents relative to construction costs. This raises both the optimal ratio (11) and the optimal density of development (12), and thereby the relative value of the undeveloped property (13).



At the optimal density of development q^* , the developed property has the net cash inflow q^*y , the resulting market value πq^*y , and the development cost $q^*\gamma$, all measured relative to the unit development cost x_1 . The developed property has the net cash inflow βy and the resulting market value $W(y)$, also measured relative to x_1 . The optimal ratio y at which to develop the undeveloped property is y^* .

Figure 1. Value of the underdeveloped property.

Table 1. Comparative statics without abandonment.

Increase in	Optimal Development Ratio y^*	Optimal Density q^*	Relative Value of Undeveloped Property $W(y)$	Restrictions on Parameters
β	+	+		
γ		—		$\frac{\delta\psi}{\delta - \beta\psi} \leq \gamma < \frac{\psi}{1 - \beta\psi}$
δ	—	+	—	$\gamma < \frac{\delta\psi}{\delta - \beta\psi}$
ι	—	—	+	$\nu_1 \geq \nu_2$
ρ	—	—	+	$\nu_1 \geq \nu_2$
σ_1^2, σ_2^2	+	+	+	$\nu_1 \geq \nu_2$

The parameters are as follows: β cash inflows from the undeveloped relative to the developed property; γ development cost of density; δ maximum density; ι riskless rate of interest; ν_i risk-adjusted, expected growth rates of the unit construction cost $i = 1$ and the unit cash inflow $i = 2$; σ_i^2 variance of the growth rates of the unit construction cost $i = 1$ and the unit cash inflow $i = 2$; and ρ correlation coefficient between the two growth rates.

These results hold if the risk-adjusted, expected growth rate for construction costs ν_1 exceeds or equals the corresponding growth rate for the operating cash inflows ν_2 .¹³ This constraint, $\nu_1 \geq \nu_2$, can be motivated as follows. Over long periods of time, construction costs affect the aggregate supply of rentable space, which, in turn, largely determines rents and thereby operating cash inflows. Because buildings are durable, over the long run rents are constrained above but not below by construction costs. Thus, the expected growth rate of cash inflows μ_2 is constrained above by the expected growth rate of construction costs μ_1 , while the variance of the growth rate of cash inflows σ_2^2 is bounded below by the corresponding variance of construction costs σ_1^2 .

When zoning restrictions are binding, the maximum density is optimal. In this case, tighter restrictions on density not only reduce the density, $q^* = \delta$, but also increase the development ratio y^* . As a result, properties are developed later on average when unit operating revenues x_2 are higher relative to unit construction costs x_1 . In this sense, restrictions on density retard development. In turn, the restrictions on density that increase (11) and decrease (12) also reduce the value of undeveloped property (13). Of course, the latter result may not hold in a housing equilibrium with endogenous rents, where tighter restrictions on density could raise rents and thereby increase the value of undeveloped property.

Unlike previous applications of option pricing, this problem has an endogenous quantity for the underlying asset. This quantity affects the optimal development ratio (11), the resulting value of the option (13), and thus the comparative statics. To see this, temporarily fix the quantity q and calculate the conditionally optimal development ratio:

$$Y(q) = \frac{\psi}{\pi} \frac{q^\gamma}{q - \beta\psi}. \quad (14)$$

The ratios (11) and (14) are equal at the optimal quantity (12): $Y(q^*) = y^*$. Also, the conditional ratio (14) is decreasing in the quantity q , $Y'(q) < 0$, for all $0 \leq q \leq \beta\gamma(\gamma - 1)$. Because the optimal quantity q^* lies in this interval, fixing the quantity q above (below) the optimal quantity q^* understates (overstates) the development ratio. That is, $Y(q) < y^*$ if and only if $q > q^*$ for all q in the above interval. Also, the relative value of the undeveloped property (13) is increasing in the optimal quantity q^* and decreasing in the optimal development ratio y^* . As a result, fixing the quantity q above (below) the optimal quantity q^* overstates (understates) the relative value of the undeveloped property. In short, with an artificially fixed quantity q , the comparative statics miss the indirect impact through the optimal density (12) of changes in parameters on both the development ratio (11) and the relative value (13).

3. Optimal abandonment

In the previous problem the undeveloped property is never abandoned because its net cash inflow is never negative. In fact, net cash inflows from undeveloped or underdeveloped properties can be negative if, for example, the rents are less than the costs of maintenance. If the current net cash inflow is sufficiently small, then it is optimal to abandon the

undeveloped property. In turn, this option to abandon a property can affect its optimal development point. The optimal abandonment and development points are computed numerically in this section.

The previous problem is modified as follows. Now the undeveloped property has the net cash inflow, $\alpha x_1 + \beta x_2$, with $\alpha < 0$ replacing βx_2 in the previous problem. The new term, the cash outflow αx_1 , represents the cost of maintenance. With this modification the valuation equation becomes

$$0 = \frac{1}{2} \sigma_1^2 x_1^2 V_{11} + \sigma_{12} x_1 x_2 V_{12} + \frac{1}{2} \sigma_2^2 x_2^2 V_{22} + \nu_1 V_1 + \nu_2 V_2 - \iota V + \alpha x_1 + \beta x_2. \quad (15)$$

This valuation equation differs from (5) only in the extra term αx_1 .

The boundary conditions are also altered. If the undeveloped property is abandoned when $x = x_a^*$, then its value is zero:

$$V(x_a^*) = 0. \quad (16)$$

Abandonment is optimal when $x = x_a^*$ only if

$$V_1(x_a^*) = 0, \quad V_2(x_a^*) = 0. \quad (17)$$

If the property is developed when $x = x_d^*$ with $q = q^*$, then its value is determined by (7), evaluated at these values. Development at x_d^* and q^* is optimal only if (8) and (9) are satisfied at these values.

The revised problem is solved as follows. Append to the previous parameters the new notation:

$$\zeta \equiv - \left(\frac{\nu_2 - \nu_1}{\omega^2} - \frac{1}{2} \right) - \sqrt{\left(\frac{\nu_2 - \nu_1}{\omega^2} + \frac{1}{2} \right)^2 + 2 \frac{\iota - \nu_1}{\omega^2}}, \quad \phi \equiv \frac{\zeta}{\zeta - 1},$$

Assuming that $\iota > \nu_1$, these parameters satisfy $\zeta < 0 < \phi$. Again using the ratio $y \equiv x_2/x_1$, define the abandonment point y_a^* and the development point y_d^* . With this new notation, the problem (6) through (10) and (15) through (17) is simplified and its solution is characterized in the Appendix. The subsequent solution is greatly simplified by considering only parameter values for which development at the maximum density, $q^* = \delta$, is optimal. Since the cost function is convex, $\gamma > 1$, a sufficient condition for development at the maximum density is $y_d^* \geq \gamma \delta^{\gamma-1}/\pi$, which can be calculated from the first-order condition for (9).

In this revised problem the optimal abandonment and development points cannot be calculated explicitly. Instead, these optimal points, y_a^* and y_d^* , can be shown to satisfy uniquely the two conditions:

$$\left(\frac{y_d^*}{y_a^*} \right)^\zeta = \frac{\psi(\alpha/\iota + \delta^\gamma) + \pi(\beta - \delta)y_d^*}{\psi\alpha/\iota + \pi\beta y_a^*} \quad (18)$$

and

$$\left(\frac{y_d^*}{y_a^*}\right)^\eta = \frac{\phi(\alpha/\iota + \delta^\gamma) + \pi(\beta - \delta)y_d^*}{\phi\alpha/\iota + \pi\beta y_a^*} \quad (19)$$

In turn, the optimal abandonment and development points affect the value of the undeveloped property. The value of this property is

$$W(y) = \frac{\alpha}{\iota} + \beta\pi y + A \left(\frac{y}{y_a^*}\right)^\zeta - B \left(\frac{y}{y_a^*}\right)^\eta, \quad (20)$$

for $y_a^* \leq y \leq y_d^*$ with

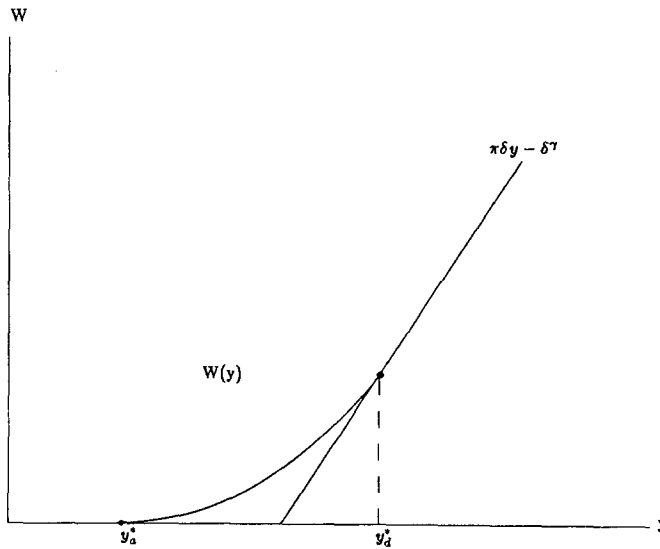
$$A \equiv \frac{(\alpha/\iota + \beta\pi y_d^*) - [\alpha/\iota + \delta^\gamma + (\beta - \delta)\pi y_d^*](y_d^*/y_a^*)^\eta}{(y_d^*/y_a^*)^\eta - (y_d^*/y_d^*)}$$

and

$$B \equiv \frac{(\alpha/\iota + \beta\pi y_d^*) - [\alpha/\iota + \delta^\gamma + (\beta - \delta)\pi y_d^*](y_d^*/y_a^*)^\zeta}{(y_d^*/y_a^*)^\eta - (y_d^*/y_d^*)}$$

This solution is depicted in Figure 2. The valuation function W is increasing and convex in the unit cash inflow relative to the unit construction cost y . As indicated, the optimal development ratio is smaller, $y_d^* < y^*$, when abandonment may be optimal, $y_a^* > 0$.

Numerical values for the optimal ratios at which the undeveloped property is abandoned and developed are presented in Table 1. With larger values of either variance, σ_1^2 or σ_2^2 , or smaller values of the correlation coefficient ρ , the optimal abandonment ratio y_a^* is smaller and the optimal development ratio y_d^* is larger. As a result, riskier undeveloped properties are abandoned later, on average. Again, risk is reduced with larger correlation coefficients ρ because the variance of the growth rate in the ratio y is $\sigma_1^2 - \rho\sigma_1\sigma_2 + \sigma_2^2$. By contrast, with larger values of either risk-adjusted growth rate, ν_1 or ν_2 , or smaller values of the riskless rate of interest ι , both the optimal abandonment and development ratios, y_a^* and y_d^* , are smaller. In this case, undeveloped properties are, on average, abandoned later but developed sooner. The remaining comparative statics are related to the costs of development. With greater costs of maintaining the undeveloped property resulting from smaller parameter values α , the abandonment ratio y_a^* is larger and the development ratio y_d^* is smaller. Thus, undeveloped properties that are more costly to maintain are on average abandoned and developed sooner. Finally, with larger values of either the cost of scale γ or the maximum density δ , both the abandonment and development ratios, y_a^* and y_d^* , are larger. In the latter case, undeveloped properties are on average abandoned sooner but developed later.



At the optimal, maximum density of development, $q^* = \delta$, the developed property has the net cash inflow δy , the resulting market value $\pi\delta y$, and the development cost δ^y , all measured relative to the unit development cost x_1 . The developed property has the net cash inflow $\alpha + by$ and the resulting market value $W(y)$, also measured relative to x_1 . The optimal ratios y at which to abandon and to develop the undeveloped property are y_a^* and y_d^* , respectively. Abandonment may be optimal, $y_a^* > 0$, because the undeveloped property is costly to maintain, $\alpha < 0$.

Figure 2. Value of the underdeveloped property with optimal abandonment.

4. Conclusion

In this paper optimal exercise policies are computed analytically and numerically for the options to develop or to abandon real estate. The results differ from standard solutions for financial options in two important details: First, the owner can determine the density or scale at which to develop his property. In part, his choice is constrained by the maximum feasible density. Second, both the operating revenues from the developed property and its cost of development evolve stochastically through time. This stochastic evolution affects the optimal date and density of development, the optimal date of abandonment, and the resulting market values of the developed and undeveloped properties. Novel results are derived for the optimal and maximum densities of development.

Notes

1. In Fischer (1978), Margrabe (1978), and Stultz (1982), the exercise price is assumed to follow a geometric Wiener process. Also, in the latter two papers the exercise price is the exchange ratio between the two risky assets. In this paper the development cost or exercise price is also driven by a geometric Wiener price, but, unlike the previous papers, the option may optimally be exercised prior to its (infinite) maturity.

Table 2. Optimal abandonment and development points.

σ_1^2	α	-.100	-.050	-.010	-.005	-.001	0
0		.095 .182	.081 .205	.044 .234	.032 .239	.014 .244	0 .245
.001		.096 .181	.081 .204	.044 .233	.032 .238	.015 .243	0 .244
.010		.094 .183	.080 .207	.043 .237	.031 .242	.014 .247	0 .248
.020		.093 .187	.078 .212	.040 .242	.029 .248	.013 .253	0 .254
.100		.080 .223	.063 .256	.024 .293	.019 .299	.007 .305	0 .306
.200		.067 .270	.049 .311	.020 .356	.013 .363	.004 .369	0 .371

δ	γ	1.01	1.10	1.25	1.50	1.75	2.00
1.01		.044 .192	.044 .192	.044 .193	.044 .193	.044 .193	.044 .194
1.10		.042 .193	.042 .195	.043 .198	.043 .203	.044 .208	.044 .214
1.25		.040 .195	.040 .199	.041 .206	.042 .219	.044 .232	.045 .246
1.50		.037 .197	.037 .205	.039 .218	.041 .243	.043 .270	.046 .299
1.75		.034 .199	.035 .210	.037 .229	.044 .270	.043 .291	.046 .353
2.00		.032 .200	.033 .214	.035 .238	.039 .284	.042 .340	.046 .406

For each value of the indicated parameters, the optimal points at which the undeveloped property is abandoned and developed are identified by the upper and lower numbers, respectively. The optimal abandonment and development points, y_a^* and y_d^* , are measured in terms of the unit cash inflow per dollar of unit construction cost, $y = x_2/x_1$.

Table 2 (continued).

ι	.010	.050	.075	.100	.150	.200
ρ						
-0.50	.002 .078	.013 .190	.021 .237	.031 .282	.051 .368	.074 .450
-0.25	.002 .088	.014 .176	.024 .222	.034 .266	.059 .350	.082 .431
0.00	.002 .076	.017 .161	.027 .206	.039 .249	.064 .332	.091 .411
0.10	.003 .071	.017 .160	.029 .200	.041 .243	.067 .324	.096 .403
0.25	.003 .064	.019 .146	.032 .190	.046 .232	.073 .313	.103 .390
0.50	.003 .052	.023 .131	.037 .173	.052 .214	.084 .292	.118 .368

ν_1	-.050	-.025	0	.025	.050	.075
ν_2						
-.050	.079 .320	.080 .303	.080 .289	.078 .277	.074 .267	.066 .258
-.025	.058 .300	.059 .281	.059 .265	.058 .251	.056 .239	.049 .229
0	.040 .283	.041 .262	.041 .243	.040 .266	.039 .211	.033 .200
.025	.024 .269	.025 .245	.025 .223	.025 .203	.024 .186	.020 .171
.050	.012 .258	.013 .231	.013 .206	.013 .182	.012 .161	.010 .143
.004	.004 .241	.004 .219	.004 .192	.004 .165	.004 .140	.003 .117

The parameters are as follows: α maintenance costs per unit of construction costs; γ development cost of density; ι riskless rate of interest; ν_i risk-adjusted, expected growth rates of the unit construction cost, $i = 1$, and the unit cash inflow, $i = 2$; σ_i^2 variance of the growth rates of the unit construction cost, $i = 1$, and the unit cash inflow, $i = 2$; and ρ correlation coefficient between the two growth rates. Unless otherwise indicated, all calculations use the parameter values: $\alpha = -.01$, $\gamma = 1.5$, $\iota = .10$, $\nu_1 = 0$, $\nu_2 = 0$, $\sigma_1^2 = .02$, $\sigma_2^2 = .10$, and $\rho = 0$.

2. Other, related papers are Anderson (1986), Arnott and Lewis (1979), Markusen and Scheffman (1978), and Mills (1981).
3. The certainty-equivalent approach in Capozza and Sick (1988) is equivalent to the argument above the valuation equation (2) in Section 1.
4. In a concurrent paper Capozza and Lee (1988) price undeveloped property and determine the optimal point and density of development. They compute an explicit solution using a risk-neutral valuation model with a Wiener process for the evolution of rent on improved land and a Cobb-Douglas production function for density.
5. Realistically, the stochastic specification (1) should be regarded as the continuous limit of a sequence with independent increments for which both the mean and variance of the growth rate is constant.
6. The model in this section is most closely related to Clarke and Reed (1987). Their risk-neutral valuation model is different in three important details. Using the notation of this paper, their endogenous density or quantity is constrained only to be nonnegative: $0 \leq q < \infty$. Their cost function $C(q)x^\gamma$ is constrained only to be increasing and convex: $C', C'' > 0$. Finally, their net cash inflow from the undeveloped property is zero: $\beta = 0$. The combination of their first and third assumptions creates a potentially serious technical problem. See footnote 13.
7. If investors could trade the state variable x_t , then the risk-adjusted mean ν_t would equal the riskless rate of interest ι .
8. A similar derivation appears in Merton (1973), Section 6.
9. Again, see the derivation in Merton (1973), Section 6.
10. For example, see Merton (1973), p. 171, or Shiriyayev (1978).
11. Rewrite the root η as follows:

$$\eta = 1 - \left(\frac{\iota - \nu_1}{\omega^2} + \frac{1}{2} \right) + \sqrt{\left(\frac{\iota - \nu_1}{\omega^2} + \frac{1}{2} \right)^2 + \frac{2}{\pi\omega^2}}.$$

Because $\pi > 0$, the parameters satisfy $\eta > 1$ and thereby $\psi > 1$.

12. To see the potential problem in Clarke and Reed (1987)—hereafter identified as CR—restrict their convex cost function C to the power function in this paper: $C(q) = q^\gamma$, with $\gamma > 1$. Also, relabel the bounds on the endogenous density or quantity q from $1 \leq q \leq \delta$, as specified in Section 2, to $q_1 \leq q \leq q_2$. With this new notation the optimal density (12) satisfies $q_1 < q^* < q_2$ if and only if $\psi q_2 / (q_2 - \beta\psi) \leq \gamma < \psi q_1 / (q_1 - \beta\psi)$. However, CR assume that $\beta = 0$. In this case, the above interval vanishes, and (12) simplifies to the solution: $q^* = q_1$ if $\gamma \geq \psi$ and $q^* = q_2$ if $\gamma < \psi$. The resulting optimal density (12) is not differentiable in the parameters, as required for the comparative statics in CR. More importantly, CR also assume that $q_1 = 0$ and $q_2 = \infty$. With these assumptions the optimal density (12) is either zero or infinite. In other words, the problem in CR is not well defined with the admissible cost function $C(q) = q^\gamma$. In CR these problems are not prominent only because a differentiable, finite optimal density q^* is assumed to exist, and an explicit solution is precluded by their general cost function C . Of course, such a solution may exist for some increasing, convex cost function C other than the power function. However, the existence of such a solution must be verified.
13. This constraint is sufficient, but not necessary. Thus, the above comparative statics must be true for all positive values of the difference, $\nu_1 - \nu_2$, that are sufficiently close to zero.

Appendix

1. VERIFICATION OF THE OPTIMALITY CONDITIONS (8) AND (9): Fix both the development point, $x = d$, and the quantity q and rewrite the value of the undeveloped property as $V(x; d, q)$. At the development point d , the boundary condition (7) must hold:

$$V(d; d, q) = P(d_2) - q^\gamma d_1. \quad (\text{A1})$$

Development at $x = d = x^*$ with the quantity $q = q^*$ is optimal if and only if

$$(x^*, q^*) = \arg \max_{d,q} V(x^*; d, q), \quad (\text{A2})$$

subject to $d \geq 0$ and $1 \leq q \leq \delta$. For the tight-fit or smooth-pasting conditions (8), differentiate (7) with respect to x , apply the first-order conditions with respect to d from (A2), and evaluate the partial derivatives at $x = x^*$. These first-order conditions must be satisfied by the optimal solution since either $d_1 = 0$ or $d_2 = 0$ is infeasible from (7) through (9) whenever $q > 0$. Finally, for (9) insert (A1) into (A2).

2. SOLUTION TO PROBLEM (5) THROUGH (10): With the transformation of variables following (10), the partial differential equation (5) simplifies to

$$0 = \frac{1}{2} \omega^2 y^2 W'' + (v_2 - v_1) y W' - (i - v_1) W + \beta y, \quad (\text{A3})$$

for all $0 \leq y \leq y_d$. Also, the initial condition (6) becomes

$$W(0) = 0. \quad (\text{A4})$$

With this transformation and (10), the boundary condition (7) becomes

$$W(y^*) = \pi q^* y^* - q^{*\gamma}. \quad (\text{A5})$$

Finally, with (10) the tight-fit condition (8) simplifies to

$$W'(y^*) = \pi q^*. \quad (\text{A6})$$

Subject to the boundary conditions (A4) and (A5), the differential equation (A3) has the unique solution:

$$W(y) = \pi \beta y + [\pi(q^* - \beta)y^* - q^*] \left(\frac{y}{y^*} \right)^\eta. \quad (\text{A7})$$

Applying (9) and (A6) to (A7) yields (11) through (13).

3. SOLUTION OF PROBLEM (6) THROUGH (10) AND (15) THROUGH (17): With the transformation of variables following (17), the partial differential equation (15) simplifies to

$$0 = \frac{1}{2} \omega^2 y^2 W'' + (v_2 - v_1) y W' - (i - v_1) W + \alpha + \beta y, \quad (\text{A8})$$

for all $0 \leq y_a^* \leq y \leq y_d^*$. At the optimal abandonment point y_a^* , the boundary conditions are

$$W(y_a^*) = 0 \quad (\text{A9})$$

and

$$W'(y_d^*) = 0, \quad (\text{A10})$$

from (16) and (17). At the optimal development point y_d^* , the boundary conditions are (A5) and (A6), evaluated at y_d^* . The general solution to (A8) is

$$W(y) = \frac{\alpha}{\iota} + \pi\beta y + Ay^\zeta - By^\eta. \quad (\text{A11})$$

Applying (A5),(A6),(A9), and (A10) to (A11) yields (18) through (20).

4. NUMERICAL PROCEDURE: For the special case with $\beta = 0$, (18) and (19) simplify to

$$y_d^* = y_d^* \left[1 + \frac{\iota}{\alpha} \delta^\gamma - \frac{\delta\iota\pi}{\alpha\psi} y_d^* \right]^{-1/\zeta} \quad (\text{A12})$$

and

$$y^* = F(y^*) \equiv \frac{\alpha\psi}{\delta\iota\pi} \left[1 + \frac{\iota}{\alpha} \delta^\gamma - \left(1 + \frac{\iota}{\alpha} \delta - \frac{\delta\iota\pi}{\alpha\phi} y_d^* \right)^{\zeta/\eta} \right], \quad (\text{A13})$$

for any $y_d^* \geq 0$. The function F is increasing and convex with $F(0) > 0 = F(\infty)$. Hence, (A13) has a unique root y_d^* , and F has the slope $-1 < F' < 0$ on $y \geq y_0$ with $F'(y_0) = -1$. See Figure 3. If $y_0 \leq y_d^*$ or, equivalently, $y_0 \leq F(y_0)$, then y_d^* can be computed as the fixed point of the contraction mapping $y_{j+1} = F(y_j)$, $j = 1, 2, \dots$, whenever $y_0 \leq y_1 \leq y_d^*$.

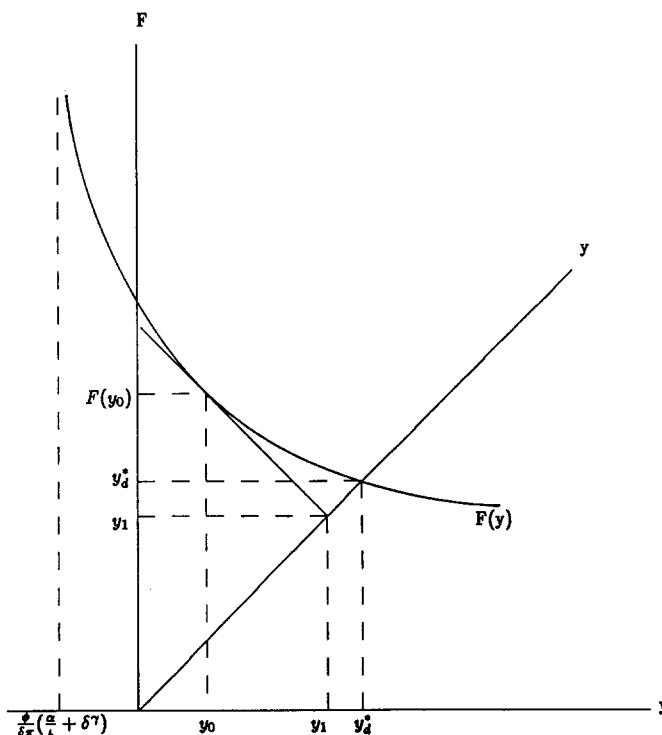
An initial value y_1 satisfying this inequality can be identified as follows. The lower bound y_0 , defined by $F'(y_0) = -1$ is

$$y_0 = \frac{\alpha\phi}{\delta\iota\pi} \left[1 + \frac{\iota}{\alpha} \delta^\gamma - \left(-\frac{\zeta-1}{\eta-1} \right)^{\eta/(\eta-\zeta)} \right].$$

Also, $y_0 \leq F(y_0)$ if and only if

$$\alpha \geq \iota\delta^\gamma \left[(1 - \zeta) \frac{\zeta + \eta}{\eta - \zeta} \left(-\frac{\zeta-1}{\eta-1} \right)^{\zeta/(\eta-\zeta)} - 1 \right]^{-1},$$

since $\zeta + \eta \leq 0$ from the previous assumption $\nu_2 - \nu_1 \leq \omega^2/2$. If this inequality holds, then y_1 is constructed such that $-1 = F'(y_0) = [y_1 - F(y_0)]/(y_1 - y_0)$ or, equivalently, $y_1 = [y_0 + F(y_0)]/2$. Again see Figure 3. This initial value y_1 satisfies $y_0 \leq y_1 \leq y_d^*$ and improves the convergence.



The function F is a contraction map on $y_0 \leq y < \infty$. The optimal development point y_d^* is the fixed point $y_d^* = F(y_d^*)$. The straight line from $(y_0, F(y_0))$ to (y_1, y_1) has the slope $F'(y_0) = -1$. This equality determines the initial value y_1 . The fixed point y_d^* is computed as the limit of $y_{j+1} = F(y_j)$, for $j = 1, 2, \dots$, as $j \rightarrow \infty$.

Figure 3. Computation of the optimal development point.

References

- Anderson, J. "Property Taxes and the Timing of Urban Land Development." *Regional Science and Urban Economics* 16 (1986), 483-92.
- Arnott, R. and Lewis, F. "The Transition of Land to Urban Use." *Journal of Political Economy* 87 (1979), 161-69.
- Capozza, D. and Li, Y. "A Generalized Model of Land Conversion under Uncertainty." Unpublished manuscript, University of British Columbia, 1989.
- Capozza, D. and G. Sick. "Risk and Return in Land Markets." Mimeo, University of Michigan, 1989.
- Clarke, H. and Reed, W. "A Stochastic Analysis of Land Development, Timing, and Property Valuation." Unpublished manuscript, University of Victoria, 1987.
- Findlay, M.C. and Howson, H. "Optimal Intertemporal Real Estate Ownership, Valuation, and Use." *American Real Estate and Urban Economics Association Journal* 3 (1975), 51-66.
- Fischer, S. "Call Option Pricing When the Exercise Price is Uncertain, and the Valuation of Bonds." *Journal of Finance* 33 (1978), 169-76.
- Margrabe, W. "The Value of an Option to Exchange One Asset for Another." *Journal of Finance* 33 (1978), 177-86.
- Markusen, J. and Scheffman, D. "The Timing of Residential Land Development: A General Equilibrium Approach." *Journal of Urban Economics* 5 (1978), 411-24.

- Merton, R. "The Rational Theory of Option Pricing." *Bell Journal of Economics and Management Science*. 4 (1973), 141–83.
- Mills, D. "The Non-Neutrality of Land Value Taxation." *National Tax Journal* 34 (1981), 125–130.
- Shiryayev, A. *Optimal Stopping Rules*. New York: Springer-Verlag, 1978.
- Shoup, D. "The Optimal Timing of Urban Land Development." *Regional Science Association Papers* 25 (1970), 33–44.
- Stultz, R. "Options on the Minimum or the Maximum of Two Risky Assets: Analysis and Applications." *Journal of Financial Economics* 10 (1982), 161–85.
- Titman, S. "Urban Land Prices under Uncertainty." *American Economic Review* 75 (1975) 505–14.